

Condensed Novelty Check: Delay–Layout Coupling in UWB Self-Calibration

Grounded in the Vicon-validated Erlangen campaign.

1 Thesis

UWB anchor self-calibration couples antenna-delay/range-bias parameters with layout scale, so same-environment tag accuracy is not a reliable proxy for physical calibration quality. In the Erlangen campaign the production AutoPos self-calibrated layout (**v4-io**) is scale-biased (metric-incorrect): an under-absorbed common-mode range bias expands it to an AutoPos-to-Vicon Sim(3) scale of 0.958 (a range-bias signature, not a true geometric distortion), while a common-mode delay re-parameterization recovers metric scale (1.010) and nearly halves the rigid anchor RMSE (105.4 to 63.0 mm). Yet the metric-correct common-mode layout gives a *worse* same-environment static median (109.5 mm) than the scale-biased **v4-io** layout (72.7 mm) when the tag delay is left uncorrected, because the layout’s scale bias compensates an uncorrected positive tag-device delay. Per-frame joint tag-delay estimation restores the common-mode layout to 72.9 mm median but destabilizes the tail (P95 317.5 mm). A scale-biased, metric-incorrect layout can therefore win in-environment *when the tag delay is left uncorrected*—once it is corrected with a single global constant the metric-correct layout becomes the better positioner (68.5 mm held-out median vs 72.7 mm; better at 17 of 24 positions, paired $p \approx 0.02$), so this is a conditional failure mode, not a claim that a metric-incorrect layout is inherently a better positioner. Mechanistically, the self-calibrated layout is internally self-consistent but not metric: the common-mode delay and isotropic layout scale span a near-degenerate direction ($\rho = -0.977$), so a family of (scale, delay) solutions fits the inter-anchor ranges equally well and range data alone cannot select the metrically true one. It is direct evidence that metric correctness and positioning accuracy must be validated separately.

2 Literature Gap

What is known	What is NOT known (our gap)
UWB self-calibration estimates anchors from radio ranges, robot motion, multihop links, or auxiliary sensing (Hamer & D’Andrea, 2018; Corbalan et al., 2023; Van Herbruggen et al., 2023; Almansa et al., 2020; Ledergerber et al., 2015; Ridolfi et al., 2021; Shi et al., 2019; Piavanini et al., 2022; Queralta et al., 2020).	These papers do not isolate whether the layout with the best tag accuracy is metrically correct, nor whether tag accuracy can rank a scale-biased layout above a validated metric one.
Antenna-delay and range-bias calibration are established UWB problems (Ledergerber & D’Andrea, 2018; De Preter et al., 2019; Shalaby et al., 2023; Liu et al., 2024; Shah et al., 2021, 2022).	In UWB ranging, delay/range-bias is usually a nuisance correction (in acoustic/TOA self-calibration it is a first-class jointly-solved unknown); its ability to absorb global layout-scale error in-environment is not characterized.
NLOS mitigation uses channel statistics, learned models, robust weighting, or biased-range models (Wymeersch et al., 2012; Di Franco et al., 2017; Marano et al., 2010; Guvenc et al., 2007; Venkatesh & Buehrer, 2007; Meghani et al., 2019; Jeong et al., 2023; Kong, 2023; Sung et al., 2023; Yang et al., 2024).	Structured positive range bias is not connected to which self-calibration geometry is physically valid.
Fisher/CRLB, rigidity, and observability analyses describe weak directions and rank deficiencies in localization and calibration networks (Shen et al., 2010; Aspnes et al., 2006; Patwari et al., 2005; Wymeersch et al., 2009; Prorok, 2013; Jourdan et al., 2008; Hesch et al., 2014; Goudar et al., 2021).	Prior work does not connect a coupled delay–scale direction to a measured scale-bias cancellation result (a scale-biased layout winning in-environment) in UWB self-calibration.
Scale-propagation-parameter ambiguity is known in acoustic self-calibration and related ranging systems (Burgess et al., 2015). AniTrack reported self-localized anchors outperforming surveyed anchors in tag accuracy (13.96 vs 16.57 cm, $n = 7$) (Luder et al., 2025).	The interaction with structured UWB range bias, and the resulting in-environment advantage of the scale-biased layout, is not reported. A consistent external observation (within one SD, competing explanations not excluded), not a diagnosis: no Sim(3) scale, delay–layout coupling, or causal intervention.

3 Our Contribution

The first contribution is scientific: we quantitatively characterize delay–layout coupling in UWB anchor self-calibration. A four-way ablation (self-calibrated vs Vicon-measured anchor coordinates $\times$ residual-correction treatments) shows the fitted residual corrections are frame-conditioned: Vicon-measured anchor coordinates degrade tag localization (311 mm RMSE with no correction, 252 mm with transplanted corrections) unless the corrections are re-estimated in the Vicon frame (78 mm). A common-mode delay re-parameterization ($d_i = c + e_i$, $c = 112$ mm) recovers metric scale from inter-anchor ranges alone (Sim(3) $0.958 \rightarrow 1.010$; rigid RMSE $105.4 \rightarrow 63.0$ mm); clamping the common mode to zero raises the solver cost by about 70%, and five perturbed initializations converge to the same scale. A gauge-free Fisher analysis on the real geometry confirms this is a strong-but-not-strict near-degeneracy: the common-mode delay correlates with the isotropic layout-scale mode at $\rho = -0.977$ (variance inflation $22\times$; profiling the layout out of the delay flattens its likelihood $22\times$; 1 mm common delay $\leftrightarrow -1.22$ mm scale at the array edge; read from the delay direction’s marginal, not the globally-softest raw eigenvector, which is a within-layer wiggle artifact). The decisive observation is that the metric-correct layout is the worse in-environment positioner *only* until the tag delay is corrected. The fair-comparison control points the same way: because the deployed firmware applies one generic antenna-delay default (16436) to every node with no per-device calibration, the physically honest correction is a single global tag-delay constant. With a leave-one-out, no-peek constant (≈ 50 mm) the metric-correct layout becomes the better positioner—68.5 mm held-out median (P95 156 mm) vs the scale-biased 72.7 mm, better at 17 of 24 positions (paired $p \approx 0.02$; the marginal-median gap alone is within noise)—and with a far more stable tail than per-frame joint estimation (which only ties the median at 72.9 mm by overfitting, P95 317.5 mm). The scale-biased layout’s apparent advantage was thus a correctable constant its scale error absorbed, not better physical calibration. An orthogonal hardware check confirms the per-device delays are not even available from the vendor: reading the factory OTP of all nine modules (the lone outlier verified by five repeated reads) shows the stored antenna delay is near-uniform (eight units 16472, one 16451) and does *not* track the fitted per-device delays—seven modules share one OTP value yet their fitted delays span ≈ 49 –148 mm—while the per-device antenna-calibration temperature is unprogrammed throughout, even though the same OTP *does* carry per-device

crystal-trim and temperature-sensor cal. Applying the OTP value is a global common-mode shift that cannot supply the per-device differences driving the coupling, so the coupling is intrinsic to commodity hardware, not a skipped calibration step.

Separately, the same campaign exhibits a *tag-side* face of the coupling: after the balanced two-layer array reaches its vertical CRLB floor, a common tag range bias aliases predominantly into height ($\text{VDOP/HDOP} \approx 1.6$), and an active vertical-injection control produced no dose-response. This is the residual-vertical limit reported by the companion system paper.

The second contribution is methodological: a Vicon-grounded falsification protocol that separates metric correctness from same-environment positioning accuracy. It uses Sim(3) anchor-scale validation, a layout/delay transfer ablation, per-anchor range-bias decomposition with error-budget closure, phase-center sensitivity, and nested spatial CV, so that a single accuracy number cannot hide the mechanism.

As a discussion-level conjecture (not a contribution), a similar failure mode could occur wherever per-device bias and geometric scale share a weakly observable direction (GNSS clock-geometry, acoustic sound-speed-geometry); the analogy is imperfect because delay is additive while scale is multiplicative, so cancellation is only approximate over the working-distance range.

A raw-frame intervention that reduces the positive tag-anchor range tail reverses the ranking in favour of the metric-correct layout; this causal intervention is the central cancellation evidence.

Scope. We claim not the bare existence of the coupling (acoustic/GNSS/AniTrack precedents preclude that) but: (i) the first gauge-free Fisher quantification of the delay $\leftrightarrow$ scale near-degeneracy on real UWB hardware ($\rho = -0.977$, $22\times$ variance inflation, 1 mm delay $\leftrightarrow -1.22$ mm edge-scale); (ii) the conditional wrong-geometry-wins regime and the control that closes it (a single global tag-delay constant makes the metric-correct layout the better positioner, 68.5 vs 72.7 mm); and (iii) the Vicon-grounded falsification protocol. The enabling three-condition combination is stated once (range-only self-calibration + independent metric ground truth for the geometry + multi-parameterization comparison), and the Fisher analysis ships as a runnable artifact so the numbers are independently reproducible. Citable: “*the common-mode antenna delay is weakly observable against isotropic layout scale ($\rho = -0.977$ on the deployed array), so same-environment accuracy can rank a scale-biased layout above a metric-correct one until a single global delay constant is applied—after which the metric-correct layout becomes the better positioner.*”

4 Nearest Prior Art

Paper	What they did	How we differ
Hamer & D’Andrea (2018)	Built a self-calibrating one-way UWB network with clock synchronization and anchor computation for multi-robot localization.	We test whether the empirically best radio-only layout is physically correct, and show that a scale-biased layout can win in-environment (under an uncorrected tag delay) by cancelling structured range bias.
Lutz et al. (2019)	Used visual-inertial SLAM and fiducial markers to estimate UWB anchor poses and sensor error-model parameters.	Their calibration is aided by external visual structure; ours diagnoses a range-only self-calibration degeneracy using Vicon only as independent validation.
Ledergerber & D’Andrea (2018)	Estimated module-dependent range inaccuracies by maximum likelihood without external ground truth.	We show that fitting delay/range inaccuracies can reward a metric-incorrect (scale-biased) solution unless anchor metric scale is checked separately.
Luder et al. (2025) (AniTrack)	Reported that self-localized DWM3000 anchors gave better tag accuracy than laser-surveyed anchors (13.96 vs 16.57 cm average error).	They report an unexplained, small-sample paradox; we propose and Vicon-test a candidate mechanism via Sim(3) scale, a four-way layout/delay ablation, and common-mode scale recovery.
Pan et al. (2015) (GNSS)	Showed satellite-clock estimation can absorb orbit error and improve PPP positioning accuracy.	Their coupled-error absorption is a GNSS clock/orbit effect; our result is a UWB self-calibrated-layout scale bias that makes the metric-incorrect layout beat the correct one when the tag bias is uncorrected.

5 Limitations

Risk	Severity	One-line mitigation
One room and 24 static positions	High	State that transfer remains unproven and freeze the pipeline for second- and N -room validation.
Results presuppose eight-anchor broadcast SS-TWR acquisition	Medium	All figures derive from the full-anchor-set broadcast acquisition the array is built for; a minimal-anchor or sequential SS-TWR scheme yields different geometry/conditioning and is not expected to reproduce them.
Cross-domain scale-delay ambiguity is prior art	Medium	Claim quantitative UWB characterization plus the in-environment cancellation result, not the bare ambiguity.
Common-mode scale recovery is a diagnostic	Medium	Present it as a mechanism re-parameterization, not as the deployed headline pipeline.
High-error tail remains	Medium	Separate median mechanism evidence from P95/dynamic performance claims.
Phase-center uncertainty	Low	Use sensitivity analysis and avoid treating the Vicon-oracle rank as the sole proof.

No prior work we found explicitly characterizes the in-environment delay–layout cancellation, though De Preter (2019), Piavanini (2022), Batstone (2017), Hamer & D’Andrea (2018) and the acoustic/GNSS work are close and are engaged directly. Target: *Measurement* (IF ~ 5.6). Working title: “Metric Correctness versus Positioning Accuracy in UWB Anchor Self-Calibration: A Vicon-Grounded Falsification Study.”

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